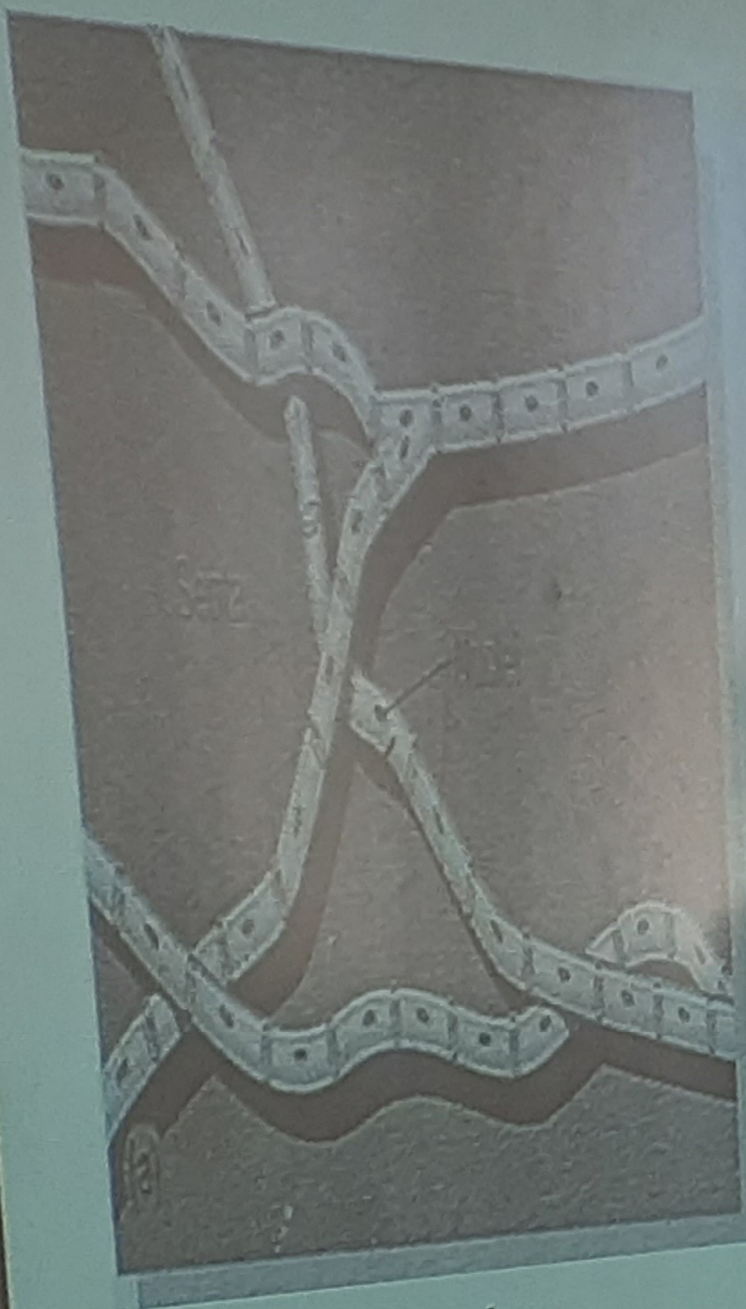


B. Moulds

- ☐ These are multicellular: form branching filaments called **hyphae**.
- ☐ A mass of interwoven hyphae is called a **mycelium**.
- ☐ The hyphae of many pathogenic moulds are:
 - **Septate**: hyphae are divided into cells by cross-walls called **septa**.
 - **Aseptate**: Hyphae without septa
 - Dermatophytes
 - *Aspergillus* species
 - Fungi causing mycetoma
 - *Penicillium marneffei*
 - Fungi causing Zygomycetes. *Mucor*, *Rhizopus*, *Basidiobolus* species

Medical important Moulds



(a) septate hyphae



(b) aseptate hyphae

C. Dimorphic fungi

□ Fungi characterized by two forms of growth.

1. **Molds:** filamentous (multicellular) fungus with **septate** hyphae in their natural reservoir or when incubated at **25°C**,

2. **Yeasts:** unicellular fungus in the **tissues** or when incubated at **37°C** on enriched media (BHI agar).

➤ Many fungal infections are opportunistic in immune-compromised, cancer patients. E.g. *Cryptococcus*

➤ *Blastomyces dermatitidis*

➤ *Paracoccidioides brasiliensis*

Histoplasma species

➤ *Coccidioides immitis*

➤ *Sporothrix schenckii*

Medical important
Dimorphic fungi

Principal mycoses (Clinical)

A. Superficial mycoses: affecting the skin, hair or nails
e.g. dermatophytosis (ringworm) and pityriasis versicolor.

- body surfaces and do not directly involve living tissues.

B. Subcutaneous mycoses: mycoses of implantation,
e.g. mycetoma, chromomycosis, subcutaneous phycomycosis, rhinosporidiosis
and sporotrichosis.

C. Systemic mycoses: deep mycoses
e.g. histoplasmosis, blastomycosis, paracoccidioidomycosis, aspergillosis and
coccidioidomycosis

- acquired by inhalation and
- May spread from the lung and involve any part of the body.

I. Direct examination/Microscopy

- Normal saline
- KOH mounts
- India Ink
- Lacto-phenol cotton blue (LPCB) mounts

II. Isolation & Culture

- SDA
- BHIA/BAP
- Media with/without antibiotics

III. Biochemical tests:

- Rapid kits for yeasts
- Urea test

IV. Special tests:

- In-vitro hair perforation test
- Germ tube test
- Chlamydoconidia formation test

1. Normal saline/ wet mount

It is useful to observe fungal elements.

Procedure

- Add the specimen to one drop of sterile physiological saline
- Apply a cover slip on the preparation
- Examine the amount under low and high power magnification.

• Interpretation

- Observe for the presence of fungal elements: budding yeast, hyphae, and pseudohyphae
- quick and simple method
- Lack of contrast, which makes identification of fungal elements somewhat difficult

2. KOH Preparation

Principle

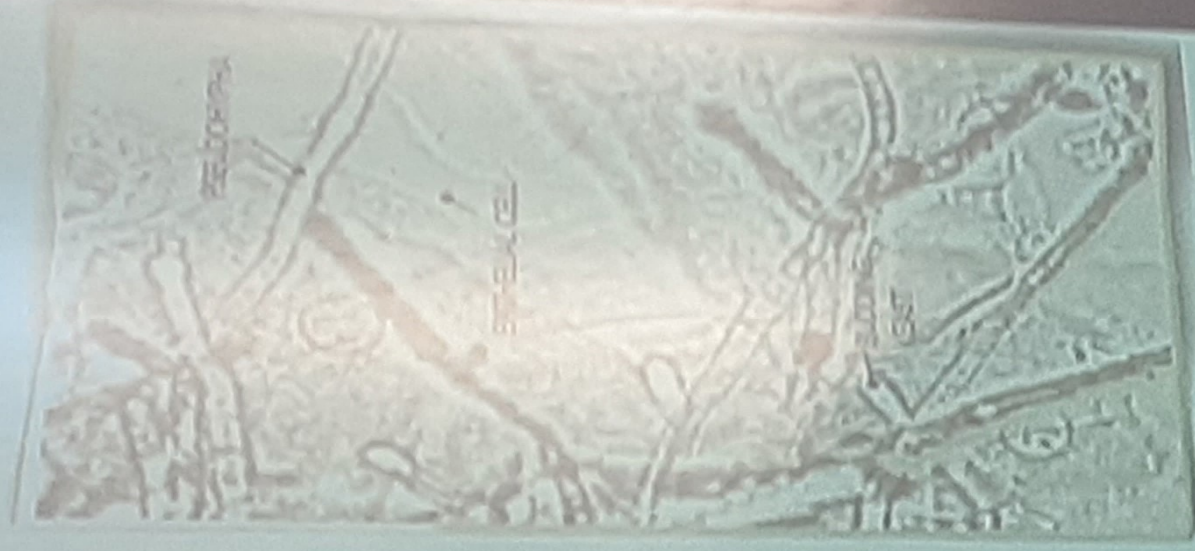
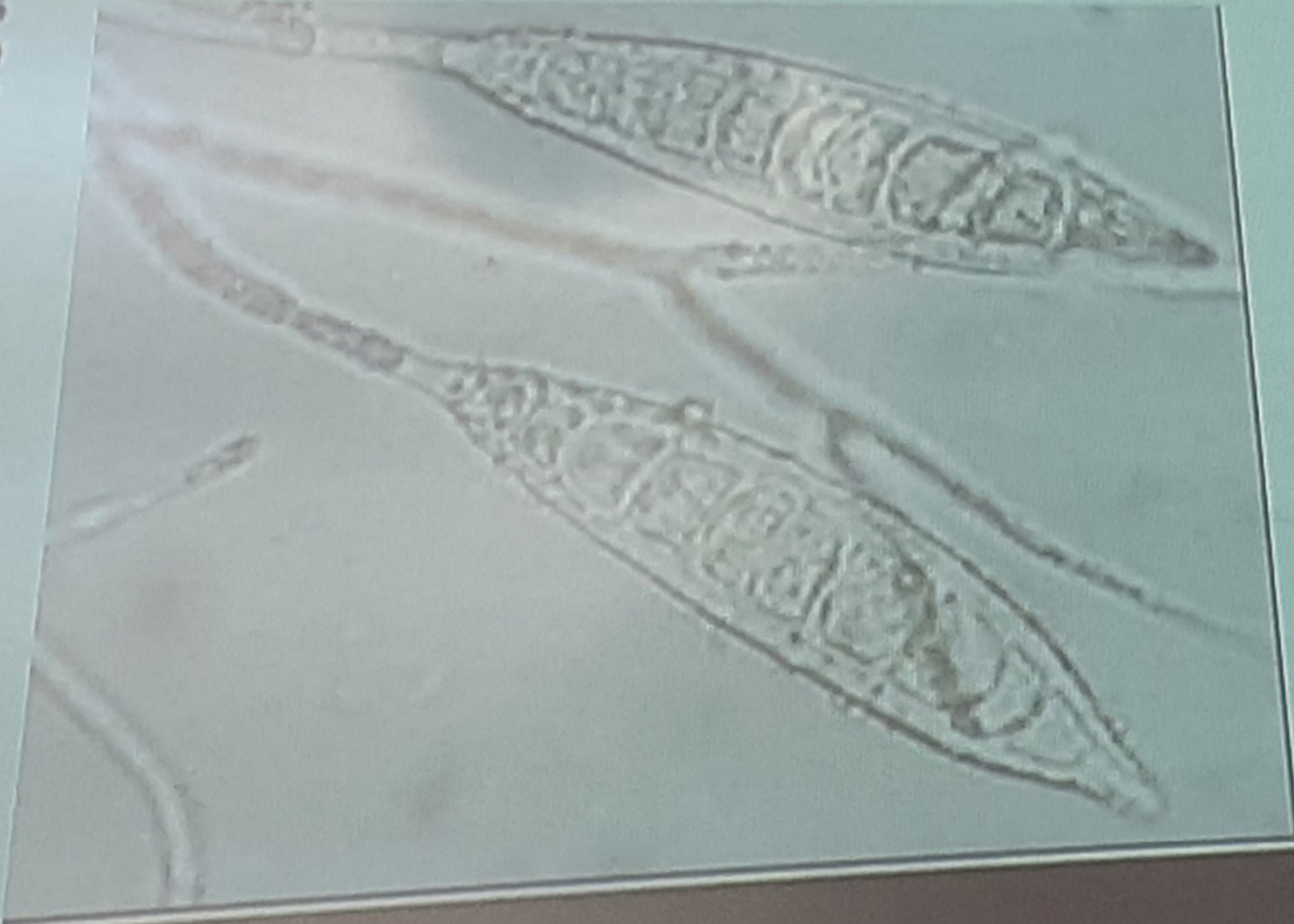
Fungal elements may be obscured by skin, hair, or nail tissue.

KOH: dissolves keratin in skin, hair or nail specimens, facilitating the observation of the organism's morphology.

Procedure

1. Place the specimen on a clean glass slide.
2. Add 1 drop of 20% KOH.
3. Place the cover glass on top of the slide
4. Place slide on the microscope stage and start with (10x) examination.
5. Examine for fungal structures at 40x

Result



Lactophenol cotton blue (LPCB)

- ❑ Used to visualize microscopic fungal morphology by imparting a blue color to the cell walls.

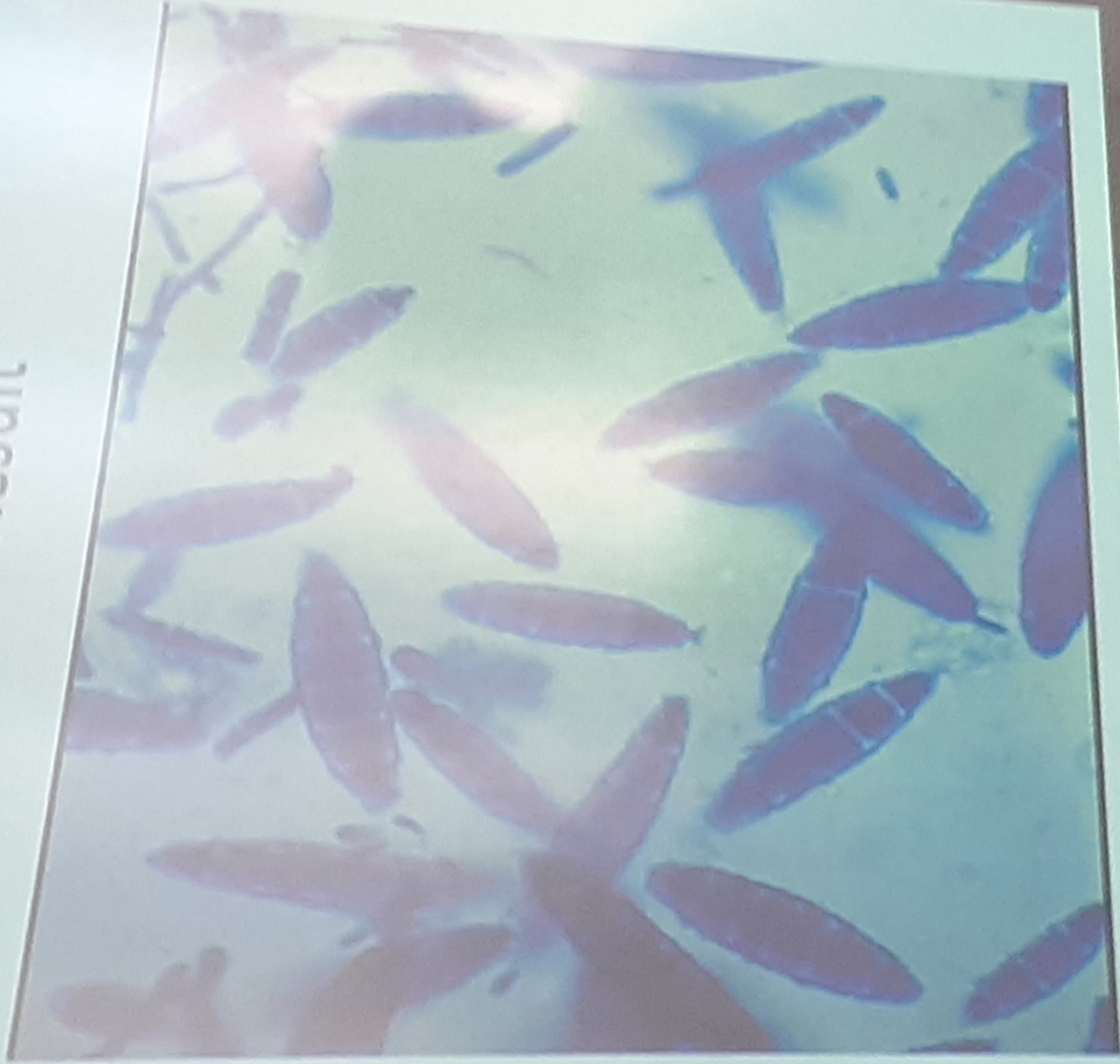
LPCB is used both as a mounting fluid & stain.

- **Lactic acid** acts as a clearing agent & aids in preserving the fungal structures.
- **Phenol** kills the organism and fixes it
- **Glycerol** prevents drying
- **Cotton blue**, an acid dye that stains the chitin present in the cell walls of fungi. It provides color to the structure.
- It can be used alone or in conjunction with KOH
- Useful to stain fungal hyphae (**molds**) & **yeasts**

Procedure

- Add One drop of LPCB to the specimen.
- Apply a cover slip.
- Examine the preparation microscopically using 10x and 40x

Result



Part Two

Sputum Sample

Collection and Acid Fast Staining (AFS)

Why we collect sputum?

- To diagnose:
 - pneumonia,
 - Bronchitis
 - TB and

Definition and suitability

- **Sputum**

- the mucous coughed up from lungs.
- not saliva or mucous from throat.
- Morning one is preferable

To be useful,

- must come from deep within the lungs.
- must be of sufficient quantity: 15ml
- TB suspected patients,
 - as many as three sputum specimens

Instruction:

- gargle with water
- Do not eat, drink, smoke or brush teeth
- Keep specimens in the refrigerator (not freezer)

Unsuitable sample

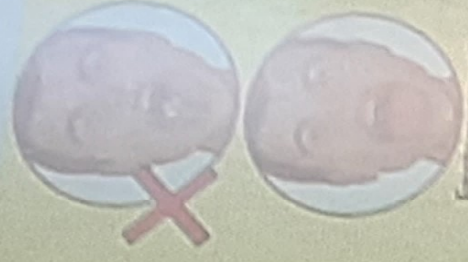
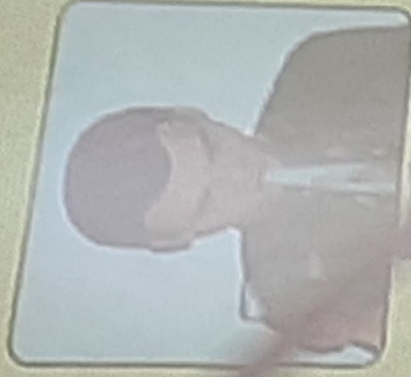
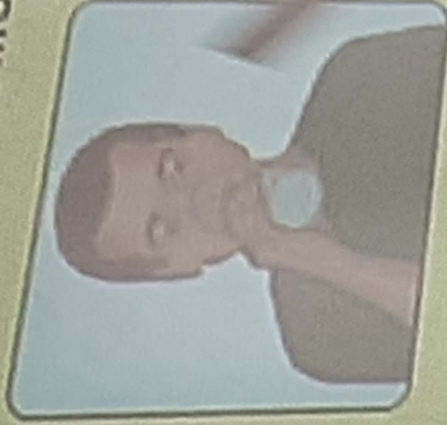
- Unlabeled specimen
- Insufficient in Quantity
- Damaged - Specimen leaked or broken in transit.
- Too Old
- Contaminated

Methods of collection

1. Collecting a sample using a cup
 - Easiest method
 - expectorate into a sterile specimen cup.
2. Nasotracheal suctioning

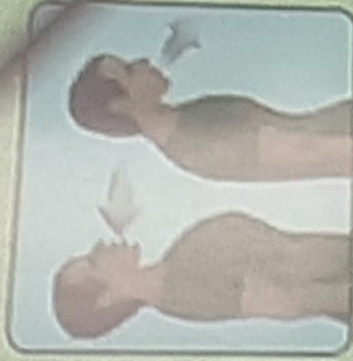
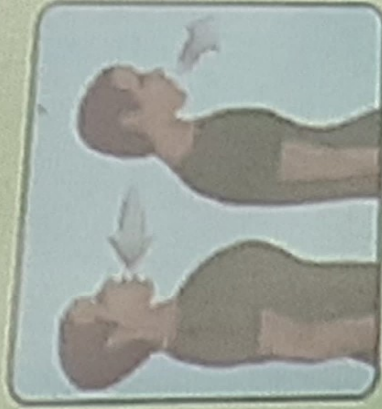
1

CLEAR YOUR MOUTH



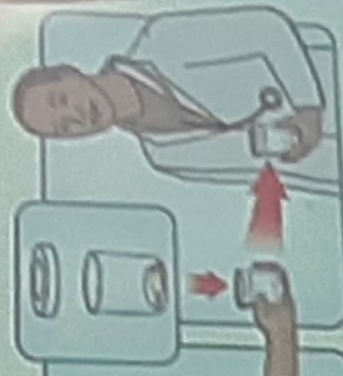
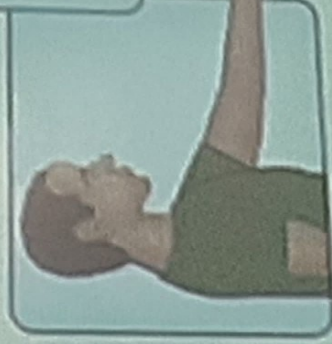
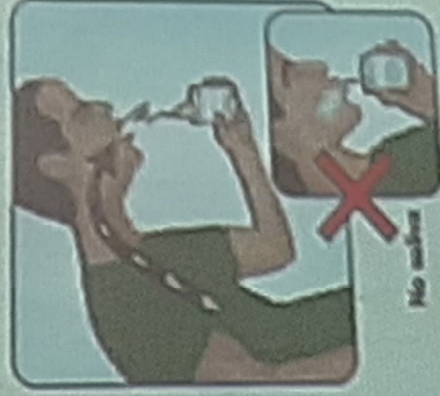
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BREATH IN AND OUT 3 TIMES



3

GIVE A SPUTUM SAMPLE



No water

Acid-Fast Staining (Ziehl-Neelsen)

- Differential staining techniques
- Franz Ziehl and Friedrich Neelsen
- Aim: to differentiate bacteria into
 - acid fast group and non-acid fast groups.
- Acid fast organisms:
 - *Mycobacterium*, *Nocardia* and *Cryptosporidium*
 - Not readily stained with simple stain

Principle:

- Carbol-fuchsin solubilizes the lipoidal material
- heat enable the reagent to penetrate lipoidal wall
- Then after all cell appears red.
- decolorized with decolorizing agent (3% HCl in 95% alcohol)

contd...

- The acid fast cells are resistant.
- The non-acid fast organisms
 - easily decolorized,
 - leaving the cells colorless
 - stain with counterstain, **methylene blue**.
- Decolorized cells
 - absorb the counter stain
 - appears blue while acid-fast cells retain color

red

Procedures

- Heat-fix the dried smear
- Cover the smear with Carbolfuchsin stain: 5 minutes
- Heat the stain until vapor just begins to rise. Do not over heat.
- Wash off the stain with clean water.
- Cover the smear with 3% v/v acid alcohol until the smear is sufficiently decolorized, i.e. pale pink for 3 minutes.
- Wash well with water.
- Cover the smear with methylene blue/malachite green stain for 2 minutes,
- Wash off the stain with clean water.
- Wipe the back of the slide clean, and place it in a draining rack for the smear to air-dry.
- Examine the smear microscopically, using the 100x oil immersion objective.

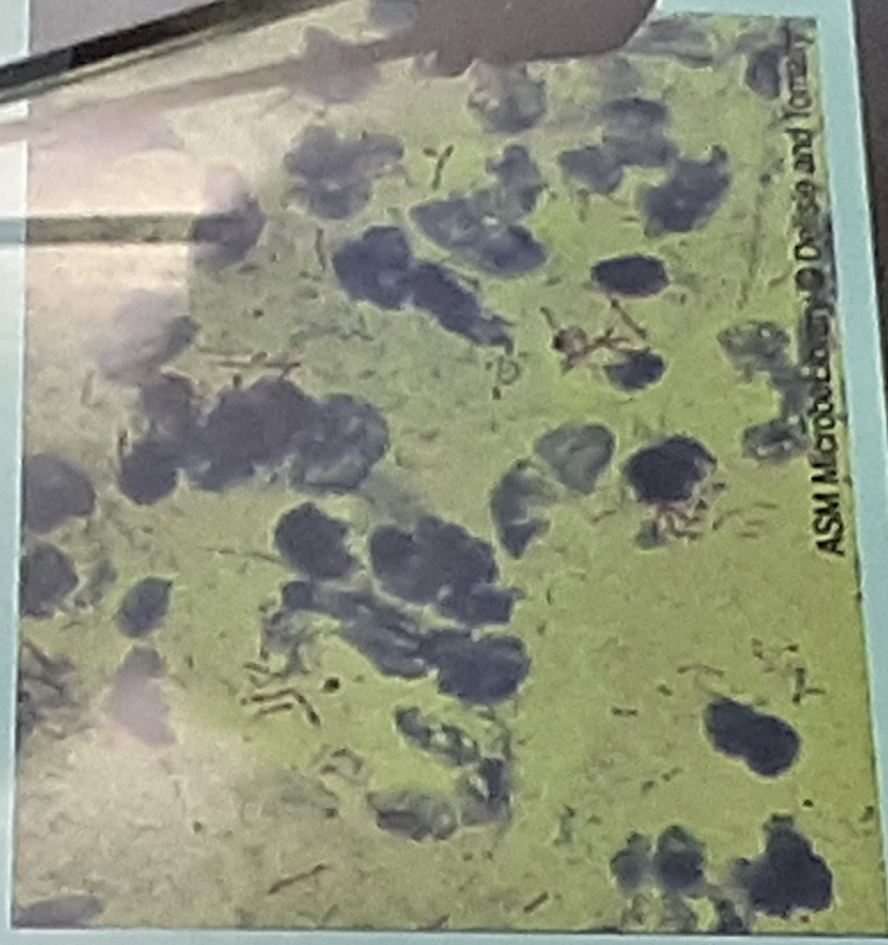
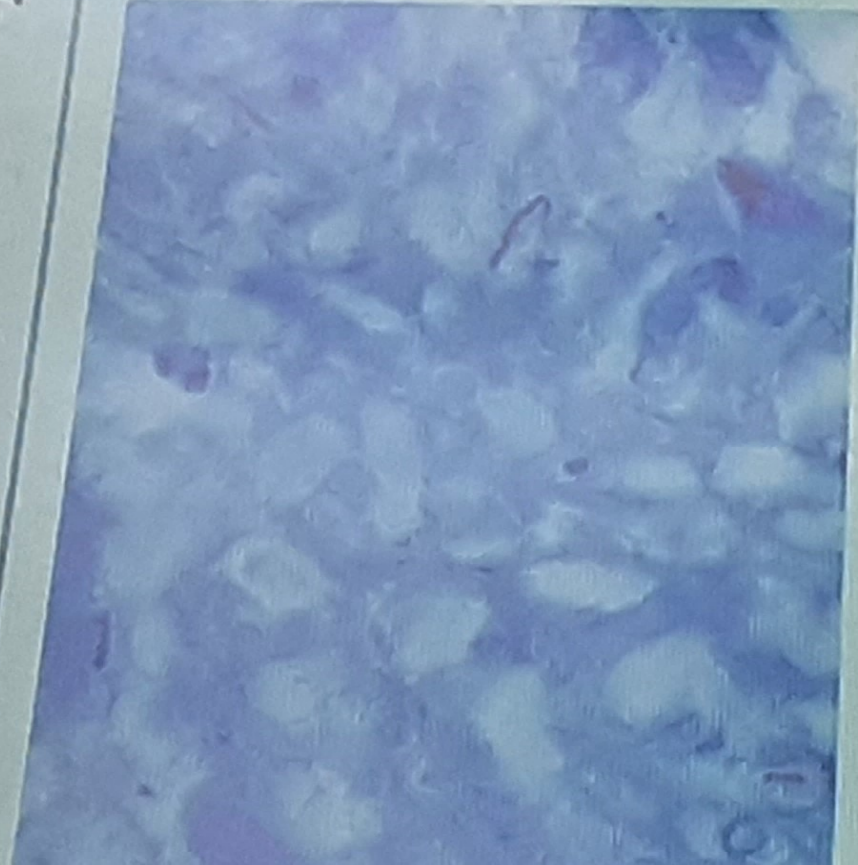
Results

- **AFB** **Red**, straight or slightly curved rods, occurring singly or in small groups,
- Cells **Blue/Green**
- Background material **Blue/Green**

Report:

- 'AFB positive'
- indication of the number of bacteria
- ❖ More than 10 AFB/field report +++
- ❖ 1-10 AFB/field report ++
- ❖ 10-100 AFB/100 fields .. report +
- ❖ 1-9 AFB/100 fields report the exact number

Ziehl-Neelsen Stain of Mycobacterium Acid-fast Rods.



Rickettsial Disease

- Rickettsiae resemble viruses in that they are ;
 - ✓ obligate intercellular parasite and
 - ✓ unable to survive as free-living organisms.
- Unlike viruses rickettsiae; like bacteria
 - ✓ contain both RNA and DNA
 - ✓ multiply by binary fission,
 - ✓ they have cell wall that contains muramic acid and enzyme.
- Based on their antigenic structure, the genus rickettsia has been divided into three main groups:
 - ◆ typhus group (*R. prowazeki*, *R. typhi*),
 - ◆ scrub typhus group (*R. tsutsugamushi*),
 - ◆ spotted fever group (*R. conori*, *R. siberica*, *R. rickettsi*).

Weil-felix reaction

A type of agglutination test most commonly used serologic test.

The reaction is based on similarity of particular antigenic determinant, which occur in most species of pathogenic rickettsia and in the;

- OX-19, OX -2 strains of *Proteus vulgaris* and
- OX-K strains of *Proteus mirabilis*.

Proteus antigen is used to detect rickettsial antibody.

This could be an example of **heterophile antigen antibody** reaction.

Weil Felix test has similar principle and procedure with Widal test.

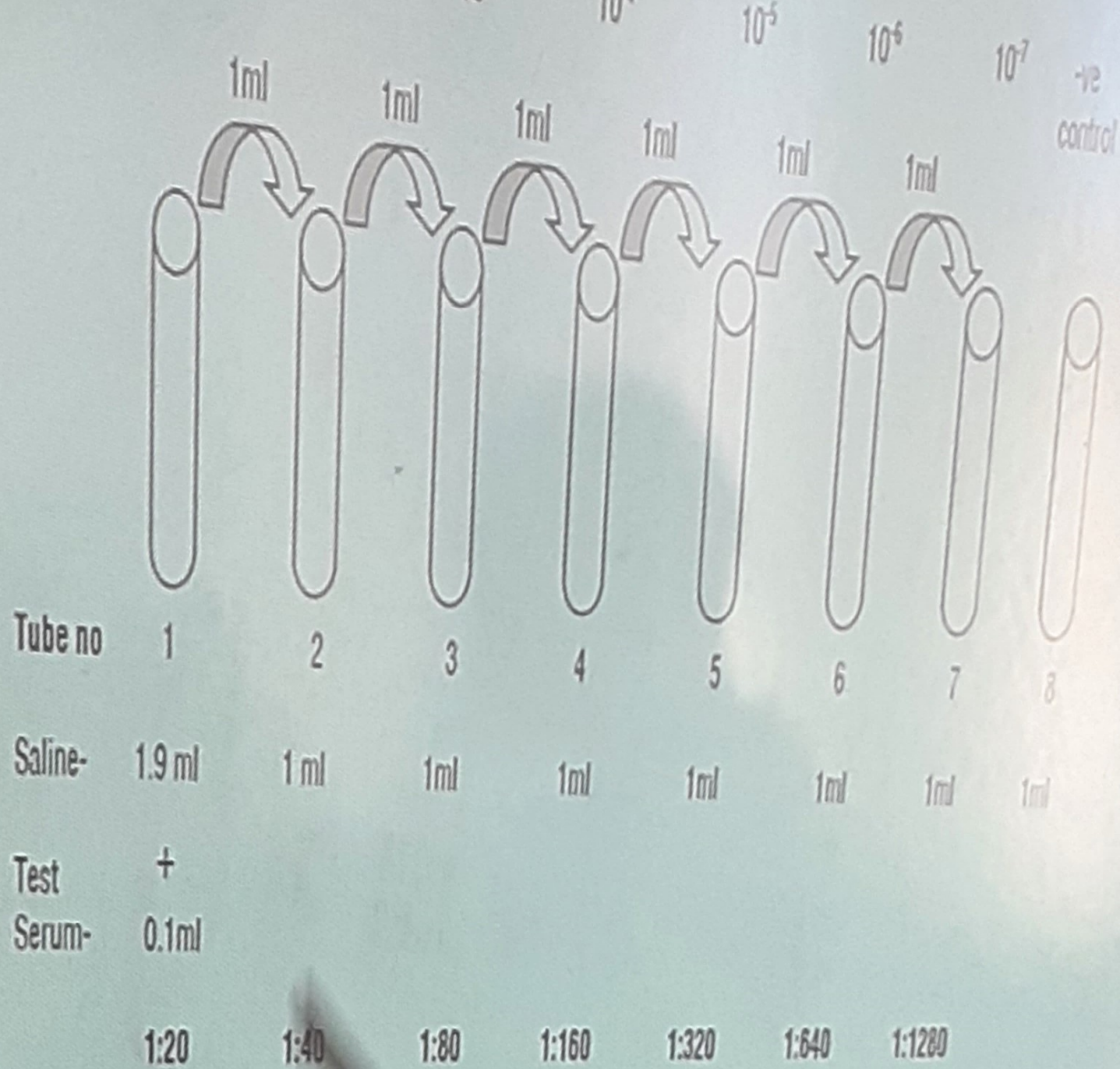
1. Rapid slide (Screening) test, procedure;

1. Place one drop of undiluted test serum on test card.
 2. Place one drop of antigen OX-19, OX -2 and OX-K in circle on test card.
 3. Mix the contents with separate applicator stick and spread to fill the whole area of the individual circle.
 4. Rotate the slide for one minute and observe for agglutination.
- If **agglutination** is **visible**, quantitative estimation of the titer of the appropriate antibodies should be done.

Procedure

2. Tube agglutination method

1. Take a set of 8 clean dry test tubes for each serum to be tested.
2. Place 1.9ml of saline in tube 1 and 1 ml of saline in other tubes (2-8)
3. Transfer 0.1 ml of undiluted serum to tube 1. Mix thoroughly. The resultant dilution of serum is 1:20.
4. Further dilutions are done in the following
 - a) Transfer 1ml of the diluted serum from tube 1 and place in tube 2 this leads to 1:40 dilutions in tube 2
 - b) Repeat the transfer process for tube 7 after mixing.
 - c) Leave 1 ml of saline in tube 8 at the 'saline control'
5. Add one drop of appropriate antigen in each
6. Mix well and incubate overnight (16-18 hrs) at 37 °C
7. Examine agglutination macroscopically.



WF reactions in Rickettsial infection

Organism	Weil-felix reactions		
	OX-19	OX-2	OX-K
Typhus Group <i>R. prowazeki</i> , <i>R. typhi</i>	+++	+/-	-
	+++	+/-	-
Scrub typhus <i>R. tsutsugamushi</i>	-	-	+++/-
Spotted fever Group <i>R. conori</i> , <i>R. siberica</i> , <i>R. rickettsi</i>	+ / +++	+++	-
	+ / +++	+++	-
	+ / +++	+++	-

+++ frequent and large antibody response

- no antibody response